Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech (CSE)

Name of the Paper: Fundamental of IoT

Semester: 3rd

Paper Code: TCS331

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	A smart home network includes devices such as smart thermostats, security cameras, lighting systems, and voice assistants. These devices communicate via an IoT platform and require efficient resource management for seamless operation. 1. Identify the clustering approach that can optimize the resource utilization in the smart home network. 2. Discuss how software agents could represent objects and synchronize data between devices. 3. Propose an identity management model that ensures secure access to smart devices while maintaining user convenience. 4. Evaluate how trust mechanisms can mitigate impersonation attacks in this smart home IoT network.	CO1 CO2
(b)	On the basis of an IoT application on e-Health. Discuss its security requirements, use cases, and potential misuse cases with flowchart of system.	CO4
(c)	1. What is clustering in IoT, and why is it important? 2. Discuss the differences between user-centric and hybrid-identity management models in IoT. 3. Discuss the challenges faced in implementing clustering techniques in large-scale IoT networks.	CO5
Q2	**(20 marks)**	
(a)	A smart city project integrates Wireless Sensor Networks (WSNs) to monitor traffic, pollution, and energy usage. Identify the key challenges in configuring and securing these networks. How can routing protocols ensure efficient data flow?	CO2
(b)	1. Discuss the structural aspects of IoT, focusing on environment characteristics, traffic characteristics, and open architecture. 2. Discuss the importance of interoperability and open architecture in IoT. How do they enhance IoT systems?	CO3
(c)	1. Define device intelligence and communication capabilities in IoT. How do these features support mobility, low-power operation, and seamless data exchange in IoT-enabled applications? 2. Discuss the role of key IoT technologies (sensor technology, RFID technology, satellite technology) in managing traffic characteristics such as latency, bandwidth, and packet loss in IoT systems.	CO6
Q3	**(20 marks)**	
(a)	A retail chain adopts RFID technology for inventory management and customer tracking. Discuss the design and technological challenges of implementing such a system. How can security challenges be mitigated?	CO2
(b)	Develop a comprehensive IoT framework that addresses critical aspects of resource management, security, scalability, and interoperability by integrating advanced clustering techniques, software agents, identity management, and key IoT technologies.	CO4
(c)	1. What are software agents, and what role do they play in IoT systems? 2. Describe how software agents facilitate object representation and data synchronization in IoT networks.	CO3
Q4	**(20 marks)**	
	Farmers use IoT devices and satellite technology for precision farming. Discuss how IoT	CO1, CO6

(a)	object identification and mobility support can enhance farming efficiency. What traffic characteristics and environmental factors should be considered?	
(b)	A remote oil rig uses satellite technology to connect IoT devices for monitoring equipment. 1. Design an IoT communication framework that supports device mobility and low power consumption. 2. Discuss how environmental factors affect traffic characteristics and propose solutions. 3. Design the flowchart of data flow in system.	CO4, CO5
(c)	1. Compare the different identity management models (Local, Network, Federated, and Global Web Identity) in IoT, highlighting their pros and cons. 2. How do user-centric, device-centric, and hybrid-identity management approaches differ in IoT systems? Provide use-case examples for each.	CO2,CO3
Q5	**(20 marks)**	
(a)	Analyze the role of IoT in autonomous vehicles. Discuss how communication capabilities and device intelligence contribute to safety and performance. Identify potential risks and propose mitigation strategies.	CO5, CO6
(b)	A city utility company is implementing IoT-enabled smart meters to track energy consumption and predict demand. 1. Propose an identity management approach that ensures secure access and protects user data. 2. Discuss how IoT scalability principles can be applied to handle a growing number of meters. 3. Evaluate the role of open architecture in supporting third-party applications.	CO3, CO4
(c)	1. Describe the architecture of Wireless Sensor Networks (WSNs). Discuss their role in IoT applications and the challenges they face. 2. Design a flowchart representing the interaction between device intelligence, communication capabilities, and mobility support in IoT.	CO1,CO2